

Malay Phadke

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🌐 [linkedin.com/in/malay-phadke](https://www.linkedin.com/in/malay-phadke)

🌐 Portfolio

Education

Sardar Patel Institute of Technology, Mumbai, Maharashtra

2021–2025

B.Tech in Electronics and Telecommunication Engineering

Minor in Computer Engineering

CGPA: 9.4/10

Department Rank: **2/152**

Relevant Coursework: Linear Algebra, Probability and Stochastic Processes, Digital Signal Processing, Artificial Intelligence and Machine Learning, Information Theory, Data Structures and Algorithms

Research Experience

Indian Institute of Science (IISc), Bengaluru

Aug 2025 – Present

Project Associate

Advisor: [Prof. Sriram Ganapathy](#)

- Worked on uncertainty estimation for multimodal language models in open-text generation.
- Studied post-training methods for improved calibration and abstention behavior.
- Contributed to benchmark design for continuous uncertainty evaluation in multimodal systems.
- Investigating overconfident hallucinations in large language models.
- Building datasets and evaluation setups for temporal reasoning in large audio language models.
- Developing mechanistic interpretability approaches for hallucination detection and uncertainty estimation in VideoLLMs.

Australian National University

Apr 2025 – Aug 2025

Research Intern (Remote)

Advisor: [Dr. Chirath Hettiarachchi](#)

- Worked on explainable reinforcement learning for glucose monitoring in Type 1 diabetes.
- Developed a post-hoc SHAP-based explainability framework to extract clinically relevant insights from reinforcement learning policies.

MITACS, University of Calgary, Calgary, Alberta

Jun 2024 – Aug 2024

Research Intern

Advisor: [Dr. Mahdis Bisheban](#)

- Explored reinforcement learning approaches for cooperative UAV-UGV path planning in search-and-rescue environments.
- Worked with ROS, Gazebo, and AirSim to study coordination and planning in simulation-based settings.
- Gained exposure to sim-to-real challenges in evaluating learning-based planning systems.

Indian Institute of Technology Bombay, Mumbai, Maharashtra

Jan 2024 – May 2024

Research Intern

- Worked on contactless biometric authentication using palmprints, deep learning, and multiresolution analysis.
- Designed a palmprint ROI extraction pipeline using MediaPipe.
- Developed deep Siamese models with Shearlet-based features for palmprint matching.

Publications

S = In Submission, * = Equal Contribution

[C.1] Debarpan Bhattacharya*, **Malay Phadke***, Sriram Ganapathy,

“BiG-SURE - Bipartite Graph Spectral energy for Uncertainty and Reliability Estimation of LLMs,”

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), (Main Conference).

[S.1] Debarpan Bhattacharya*, **Malay Phadke***, Md Sahil Ansari, Sriram Ganapathy,

“Multimodal Open-text Uncertainty Estimation for Large Language Models,”

Under Submission at ACM Transactions on Intelligent Systems and Technology

[S.2] **Malay Phadke***, Akshay Gokhale*, Om Patil, Neel Patel, Preyes Parab, Sujata Kulkarni, and Kiran Talele

“Multi-Agent Reinforcement Learning for Path Planning: A Structured Survey Across UAV, UGV, and Hybrid Robotic Systems,”

Under Submission at IEEE Transactions on Emerging Topics in Computational Intelligence

[J.1] S. Kulkarni, **M. Phadke**, A. Sawant, N. Patel, O. Patil,

“Advancements in Seismic Data Collection and Analysis Through Machine Learning,”

Industry Experience

TANUH, Bangalore, Karnataka

July 2026 – Present

Engineering Consultant

- Working on AI models for oral cancer screening.

Vista IoT, Mumbai, Maharashtra

Jan 2024 – May 2024

Project Lead

- Led a consulting project to develop an IoT gateway for industrial applications.
- Implemented communication support for protocols including MODBUS, OPC-UA, and IEC standards.

Drona Aviation, Mumbai, Maharashtra

Jun 2023 – Sep 2023

Software Developer

- Developed Python and C++ ROS packages for nano-drone control and camera streaming.
- Built gesture-control and object-tracking modules and supported migration from ROS1 to ROS2.
- Optimized flight-control firmware to improve latency and stability.
- Worked on remote operation of a hardware drone over the internet using the MultiWii protocol.

Patents

NAZAR – ML-based Solution for Surveillance Security

Indian Patent Office – Application No: 202421053039

UAV–UGV Collaboration for Port Monitoring

Indian Patent Office – Application No: 202421073880

Achievements

- Selected for MITACS Globalink Research Internship at University of Calgary, Alberta, Canada (Top 3000 / 40,000+ applicants).
- Awarded Academic Excellence Award for outstanding undergraduate performance at SPIT.
- Ranked 1st in the Electronics and Telecommunication Department for three consecutive academic years.
- Won Kavach Hackathon (National Cybersecurity Challenge, 3900+ teams) for cryptocurrency analysis.

Selected Projects

CLARIS: Cluster Access and Job Management Platform

FastAPI, Next.js, Docker

- Built a web platform for unified access to multiple GPU clusters within the IISc network.
- Developed a FastAPI backend and Next.js frontend to support job submission, log and file inspection, terminal access, and cluster management through a single interface.
- Used Docker-based user sessions to maintain isolated connection state for each login and integrated an LLM-based assistant for log inspection and issue resolution.

Drones for Smart Agriculture (IEEE AESS Global Sponsored Project)

- UAV–UGV collaboration system funded \$25k by IEEE AESS Global under DSTEI scheme.
- Designed an end-to-end pipeline for crop-health classification comparing EfficientNet, MobileNet, ResNet, and YOLOv7 along with classical image preprocessing algorithms.
- Deployed on NVIDIA Jetson Orin Nano for real-time inference.

Certifications

- Coursera – DeepLearning.AI TensorFlow Professional Certificate
- Codecademy – Python, C/C++, Go, Ruby
- NPTEL – Computer Vision and Image Processing (Topper)
- NPTEL – Introduction to Embedded Systems (Topper)
- TOEFL iBT – 111 / 120 (R28, L28, S27, W28)

Tools & Technologies

Python, C/C++, TensorFlow, PyTorch, MATLAB, ROS/ROS2, AirSim